

Dexter Fernandes

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https://github.com/Dexter-Fernandes

AI Engineer

Applied AI Engineer with 3+ years building production Python-based services, now focused on AI agent infrastructure, RAG, and LLM evaluation. Built a stateful LangGraph agent orchestrating retrieval, correctness, and grounding tools through conditional workflows, exposed via typed MCP tools for diagnosis. Experienced with relational databases (PostgreSQL), backend API design, and prompt engineering for production LLM systems.

PROJECTS

FinanceBench RAG Evaluation Harness

Jun 2026

- Built a **RAG evaluation harness** for financial-document question answering to diagnose whether failures stem from retrieval, generation, or grounding.
- Built a **hybrid-retrieval** pipeline combining **BM25**, **FAISS**-backed dense embeddings, and **reciprocal rank fusion** with **cross-encoder reranking**, with document chunking via **recursive character text splitting**, and three isolable retrieval modes (closed-book, oracle-context, dense) to separate retrieval quality from generation quality.
- Built a stateful **LangGraph** agent orchestrating retrieval, correctness, and grounding tools through conditional workflows to classify RAG failures, with structured Pydantic outputs benchmarked for diagnostic accuracy, latency, and tool usage against a deterministic baseline.
- Designed a dual-track evaluation layer pairing deterministic checks (exact/normalised match, scale-aware numeric matching) with **LLM-as-judge** scoring, joined against retrieval metrics to attribute wrong answers to retrieval, generation, or grounding failure.
- Benchmarked the pipeline against **Snowflake Cortex Search** on identical chunks and metrics to validate results against a production-grade retrieval baseline.
- Exposed runs, results, and **failure diagnostics** through typed MCP (Model Context Protocol) tools for programmatic and human-readable inspection.

WORK EXPERIENCE

Independent Projects & Upskilling

Sep 2024 - Present

Self-directed

- Built a stateful LangGraph agent orchestrating retrieval, correctness, and grounding tools through conditional workflows to classify RAG failures, benchmarking diagnostic accuracy, latency, and tool usage against a deterministic baseline.
- Completed Nvidia "Building RAG Agents with LLMs" (Jul 2026), covering RAG, agents, and evaluation with LangChain and LangSmith.

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, with PostgreSQL for structured pipeline and metadata storage, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.

- Presented model performance, failure modes, and deployment trade-offs to engineering teams, client stakeholders, and field operators, turning messy production signals into clear optimisation recommendations.
- Built internal web tooling using Django and Bootstrap to support pipeline operations and reporting.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries and John Deere.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% on large real-world datasets.
- Worked with hardware, software, QA, and client teams to translate model performance, latency, and failure analysis into rollout decisions for deployed safety systems.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

CERTIFICATIONS

Building RAG Agents with LLMs

Nvidia

Jul 2026

Deep Learning Specialisation

deeplearning.ai

Jul 2019

SKILLS

- **AI-Assisted Development:** Claude Code, ChatGPT Codex
- **RAG and Retrieval:** document-grounded QA, retrieval metrics, RAG evaluation, RAG pipelines, dense retrieval, hybrid retrieval, FAISS vector store, ChromaDB, BM25, reciprocal rank fusion, cross-encoder reranking
- **Agent Engineering and Evaluation:** LLM-as-judge, agent orchestration, tool orchestration, stateful agent workflows, MCP, prompt engineering, evidence matching, citation checks, hallucination detection, error analysis, structured outputs, Pydantic (v2), FastAPI
- **Experimentation and Reliability:** failure-mode analysis, benchmarking, regression checks, hypothesis testing, ablation studies, post-deployment validation, production monitoring
- **Python and ML Engineering:** Python, Pandas, NumPy, scikit-learn, PyTorch, MLflow, DVC, Apache Airflow, Optuna, LangChain, LangSmith, PyDantic, FastAPI, Snowflake, LangGraph
- **Production Systems:** Docker, CI/CD, Git, structured logging, observability, health checks, Apache Kafka, Linux, schema-validated output contracts, Microsoft Azure, SQL, PostgreSQL, Django, Bootstrap